

Non-linear operation
that correlates with shape
on morphology of features

Morphological Processing

applied on Binary Image

Morphological processing pursues the goal of removing imperfections that arise when converting a greyscale image to a binary image

Defn: A collection of non-linear operations related to the shape or morphology of features in an image.

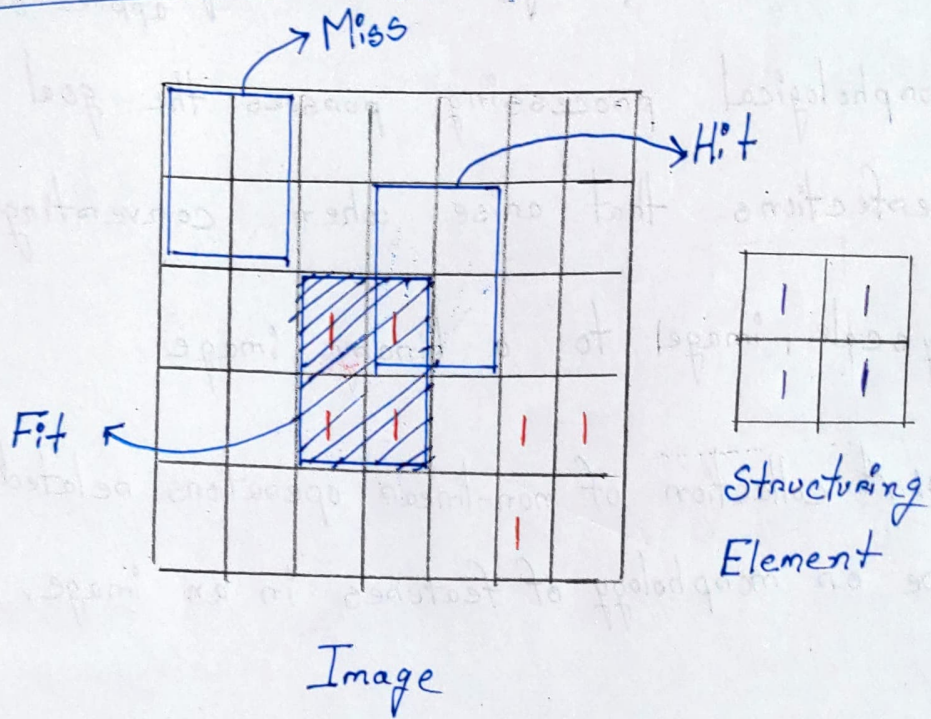
Structuring Element depends on number of 1s

Morphological techniques probe an image with a small shape/template (a small matrix of pixels each with a value of zero or one) called a Structuring Element

Matrix Dimensions = Size of the Structuring Element

Pattern of Ones/Zeros = Shape of the Structuring Element

Fit, Hit & Miss:



depends on number of 1s

Structuring Element

Morphological techniques probe an image with a small

shape/template (a small matrix of pixels each with a

value of zero or one) called a Structuring Element

Matrix Dimensions = Size of the Structuring Element

Pattern of Ones/Zeros = Shape of the Structuring Element

Calculating Origin of Structuring Element:

Consider (1,1) as first position of structuring element

Put values of ^{size} last coordinate in formula

0,0		
	1,1	
		2,2

3x3

Formula [Center = (x,y)]
6x5 structuring element

$$x = \text{Floor} [(6+1)/2]$$

$$= 3$$

$$y = \text{Floor} [(5+1)/2]$$

$$= 3$$

	1	2	3	4	5
1					
2					
3			①		
4					
5					
6					

6x5

[Mention in Exam which value you've used]

⊗ Mention starting point used (0,0) or (1,1) ✓

Mention Structuring Element Size

$$x = \text{Floor} [(\text{Number of Rows} + 1) / 2]$$

$$y = \text{Floor} [(\text{Number of Column} + 1) / 2]$$

Show Like This In Exam



Dilation & Erosion:



Dilation $\Rightarrow$ A fundamental morphological operation that increases foreground depending on a structuring element $f \oplus s$



Erosion $\Rightarrow$ A fundamental morphological operation that shrinks/removes foreground depending on a structuring element $f \ominus s$

(objects larger than S.E.) $\swarrow$ structuring element $\searrow$ (objects smaller than S.E.)

Steps

- ① Calculate Center of Origin of Structuring Element (Draw SE on plain paper)
- ② Padding according to Structuring Element & its center
- ③ Observe Fits, Hits and Misses

Erosion

The erosion of a binary image f , by a structuring element s (denoted by $f \ominus s$) produces a new binary image $g = f \ominus s$ with ones in all locations (x, y) of a structuring element's origin at which that structuring element s fits the input image f .

$$\text{i.e. } g(x, y) = 1 \text{ if } s \text{ fits } f$$

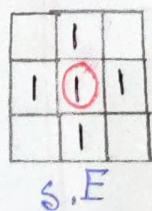
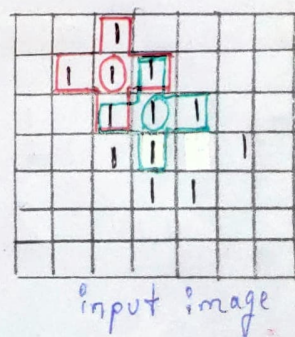
0 otherwise

Considering
background $\rightarrow 0$ (black)
foreground $\rightarrow 1$ (white)

repeating for all pixel coordinates (x, y)

Shortcut:

- ① In input image, find how many S.E. present inside
- ② For each S.E. present in the input image, place 1 in center of origin
- ③ All other pixels will be 0.



Draw Output Image & Structuring Element only
 (Only Draw 1's)
 No need to draw 0's

Erosion with 3x3 Structuring Element

Two techniques

① Only keep 1's (in center pixel) where the Structuring Element fits

② Remove all boundary 1's

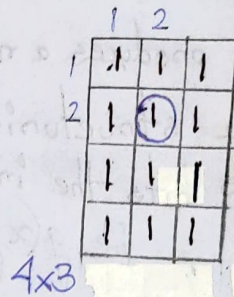
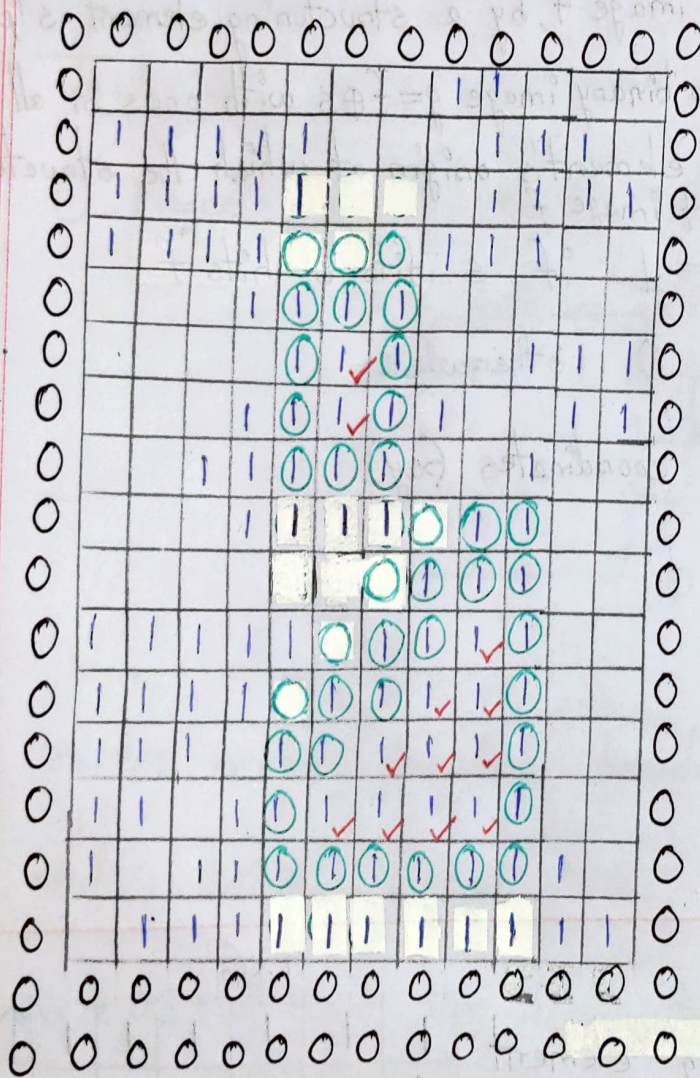
Only for
3x3

1 1 1
1 1 1
1 1 1

S.E

0 0	1 1 1	0	
0	1 1	1 1	0
1 1	1 1	1 0	⇒ 1 1
1 1	1 1	1 0	1 1
0	1 1 1 1		

Opening Practice



Center of origin (x, y)

$$y = \text{floor}[(4+1)/2] = 2$$

$$x = \text{floor}[(3+1)/2] = 1$$

$$(x, y) = (2, 2)$$

Erosion ✓

Dilation ○

Dilation

The dilation of a binary image f , by a structuring element s (denoted by $f \oplus s$) produces a new binary image $g = f \oplus s$ with ones in all locations (x, y) of a structuring element's origin at which the structuring element s hits the input image f .

$$\text{ie } g(x, y) = 1 \text{ if } s \text{ hits } f$$

0 otherwise

repeating for all pixel coordinates (x, y)

Shortcut:

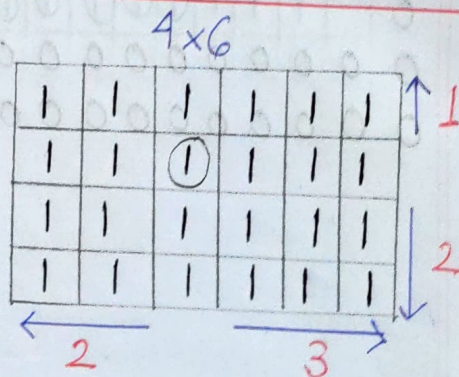
4x6 structuring element

In input image,

For every 1

① Fillup 2 rows above & 1 row below

② Fillup 3 columns left & 2 columns right



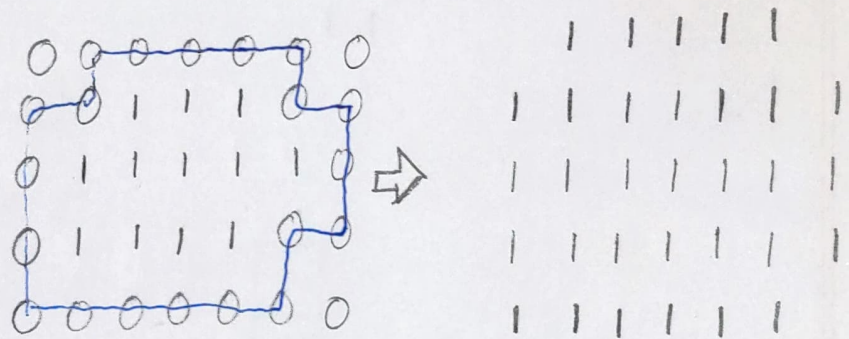
$3 \times 3 \Rightarrow$ Fill up / Increase one pixels around boundary

$5 \times 5 \Rightarrow$ Fill up / Increase two pixels around boundary

Dilation with 3×3 Structuring Element

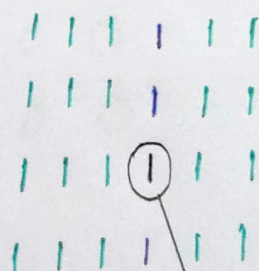
Technique

① Place 1's around boundary



Note: When drawing boundary, use only horizontal and vertical lines, NOT diagonal lines

Note 02: Draw boundary twice if S.E is 5×5



1 in input
image

• $\rightarrow$ 3 columns left,
2 columns right

• $\rightarrow$ 2 rows above,
1 row below

Opening & Closing

Opening $\Rightarrow$ Erosion followed by Dilation $f \circ s = (f \ominus s) \oplus s$

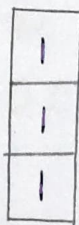
Closing $\Rightarrow$ Dilation followed by Erosion $f \bullet s = (f \oplus s) \ominus s$

Practice (Opening)

```

0 0 0 0 0 0
0 0 1 1 0 0
0 1 1 1 1 0
0 0 1 1 0 0
0 0 0 0 0 0
    
```

image



structuring
element

$$g = (f \ominus s_e) \oplus s_d$$

Mention
formula
in exam

* this part must
be written in exam

1) Erosion

```

0 0 0 0 0 0
0 0 0 0 0 0
0 0 1 1 0 0
0 0 0 0 0 0
0 0 0 0 0 0
    
```

Eroded
Image

2) Dilation of Eroded Image

```

0 0 0 0 0 0
0 0 1 1 0 0
0 0 1 1 0 0
0 0 1 1 0 0
0 0 0 0 0 0
    
```

Final Result
after Morphological Opening

Practice (Closing)

0 0 0 0 0 0
 0 0 1 1 0 0
 0 1 1 1 1 0
 0 0 1 1 0 0
 0 0 0 0 0 0
 image

structuring element

$$g = (f \oplus s) \ominus s$$

① Dilation

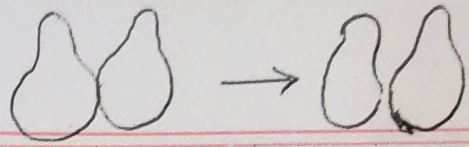
0 0 1 1 0 0
 0 1 1 1 1 0
 0 1 1 1 1 0
 0 1 1 1 1 0
 0 0 1 1 0 0

Dilated Image

② Erosion of Dilated Image

0 0 0 0 0 0
 0 0 1 1 0 0
 0 1 1 1 1 0
 0 0 1 1 0 0
 0 0 0 0 0 0

Final Result after Morphological Closing

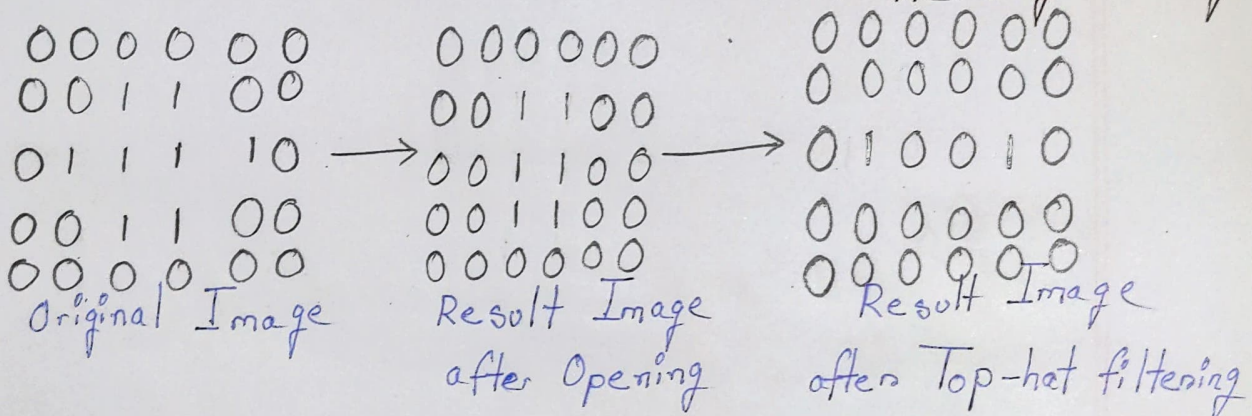


Two types of Morphological Filtering

① Top-hat filtering

Performed on grayscale/binary image and returns a filtered image

Computes the morphological opening of the image and then subtracts the result from the original image



② Bottom-hat filtering

Performed on grayscale/binary image and returns a filtered image

Computes the morphological closing of the image and then subtracts the original image from the result



Segmentation [Last step of Pre-Processing]

Refers to the process of partitioning a digital image into multiple segments based on certain criteria or features

Each segment typically represents a set of pixels having similarities in:-

- intensity
- shape
- color
- texture

Note:

Only in Binary Image, shape of objects considered (usually)

Applications of Segmentation

① Object Recognition and Detection

② Image Compression

③ Image Editing and Enhancement

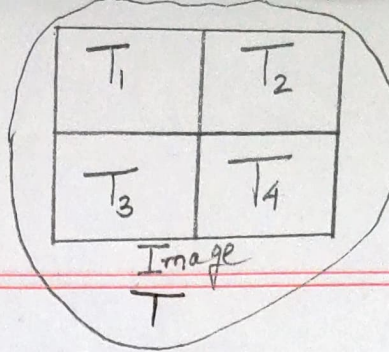
④ Medical Imaging

⑤ Remote Sensing

☐ Types of Segmentations

① Thresholding

> Threshold value chosen



Local threshold $\Rightarrow T_1, T_2$

Global threshold $\Rightarrow T$

Image Binarization

One global threshold value T applied

$< T \Rightarrow$ One class (0 or Black)

$\geq T \Rightarrow$ Another class (1 or White)

Threshold Selection

Most frequently employed method for determining threshold value is based on histogram analysis of intensity levels.

Problems

- ① Valley may be too broad, making it difficult to locate a significant minimum
- ② Multiple valleys, multiple minima
- ③ No visible valley
- ④ Noise

Notes

Histogram Plotting cannot be done for color image

Color image must be split into its 3 channels, and separate binary images must be created then merged.

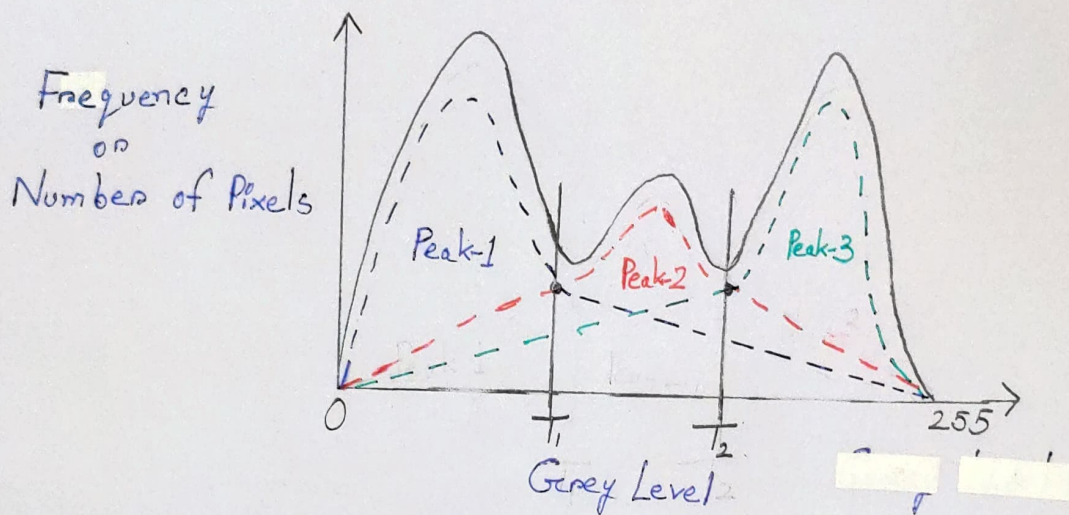
Over Segmentation

- Required regions removed on feature
- Threshold value too low

Under Segmentation

- Unwanted regions still present on features
- Threshold value too high.

Optimal Threshold Value Selection



- For each peak, draw a curve as shown above from minimum (0) to maximum (255) grey level value
- Each intersection between the curves (of peaks situated next to one another, for example: Peaks 1 and 2 or Peaks 2 and 3) is a potential optimal threshold
- If multiple optimal thresholds found, each must be applied to observe whether over-segmentation or under-segmentation occurs.